
On Reliability of Efficient Membership Inference Vulnerability Evaluation

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Abstract

Membership inference attacks (MIAs) are popular methods for empirically assessing the leakage of sensitive information in the training data through models or statistics learned from the data. The MIA vulnerability is often evaluated through false positive rate (FPR) and true positive rate (TPR) of a binary classifier that tries to predict whether a particular sample was in the training data. However, in order to reliably estimate the TPR especially for low FPR values, a lot of observations are needed, which in case of MIA translates to many target models, leading to large computational cost. To avoid excessive compute requirements, the MIA scores are often averaged over multiple individuals and multiple targeted models. We demonstrate two key weaknesses in this efficient MIA evaluation pipeline. First, we show that evaluating the TPR based on MIA scores concatenated across multiple individuals, commonly used to study vulnerabilities in the very low FPR regime, is not calibrated across the per-sample FPRs. This makes it unreliable as a tool for auditing differential privacy. To solve this, we propose a post-processing method to effectively calibrate the FPR across different samples. Second, we identify a finite population bias in the commonly used efficient likelihood-ratio attack (LiRA) implementation proposed by Carlini et al. (2022), leading to a positive bias in the per-sample vulnerability.

1 Introduction

Efficiently evaluating the privacy risk associated with a particular machine learning algorithm applied to a particular dataset is critical for calibrating the level of privacy protection, but no good general solutions exist for this problem. The difficulty of the problem arises from many sources: 1) the privacy risk is a statistical phenomenon and evaluating it requires repeated observations; 2) identifying the most vulnerable samples needed to evaluate maximum risk can be difficult; and 3) it is difficult to know if the applied attack is optimal.

Privacy auditing methods (Annamalai et al., 2026) address each of these by 1) developing specific one-run-auditing (ORA) techniques (Steinke et al., 2023; Mahloujifar et al., 2025); 2) crafting canary samples to represent maximally vulnerable samples in the entire data universe (Boglioni et al., 2026); and 3) potentially using more easily auditable internals of the algorithm to ensure strong attacks. These methods have weaknesses: the reliability of ORA is fundamentally limited (Xiang et al., 2025; Keinan et al., 2025), and the actual samples may be more or less vulnerable than the canaries.

To evaluate the vulnerability of the actual samples, we focus on membership inference attacks (MIAs) for privacy risk estimation. MIA is based on a binary classification problem: was a given sample part of the training dataset or not. MIAs are useful for this purpose because unlike stronger privacy attacks like attribute inference and training data reconstruction, the setting is highly standardised.

MIA vulnerability is related to vulnerability to these other attacks by reductions (Salem et al., 2023). The most widely used privacy enhancing technology in machine learning, differential privacy (Dwork et al., 2006), can be naturally interpreted as a bound on MIA success (Wasserman and Zhou, 2010).

Following Carlini et al. (2022), MIA evaluation has focused on evaluating the true positive rate (TPR) of membership identification at fixed low false positive rate (FPR), as this is most sensitive to the adversary’s ability to confidently detect even a small number of true members of the training dataset. The downside of the focus on low FPR values is that estimating the corresponding TPR values reliably requires a lot of membership data.

Given M attack target models with N attack target points each, a per-data-point evaluation requires at minimum $\text{FPR} > 1/M$ and for reliable estimation $\text{FPR} \geq 10/M$ is advisable. Reaching even moderately low $\text{FPR} \leq 0.1$ in this setting requires $M \geq 100$ which is often impractical.

In order to increase the number of target evaluations, it is appealing to mix evaluations from different target models and attacks for a total of MN target evaluations. This approach was proposed by Carlini et al. (2022) for efficient MIA vulnerability evaluation and is also used by the ML Privacy Meter library (Murakonda and Shokri, 2020). This approach allows formally having enough evaluations for a very low FPR, but unless one is very careful, the interpretation of these results can be difficult. In particular, we show that the global FPR from this process is usually not what would be expected, which can lead to misleading evaluations of the privacy risk, and propose a correction. We also highlight a finite population bias in this estimation process and propose a fix.

Contributions

- We study per-sample and concatenated MIA evaluations under binary hypothesis test framework. We explicate the different sources of randomness across the two aggregation methods, making it easier to understand the difference in the vulnerabilities they report (Section 3).
- We show that concatenated MIA aggregation reports an average over per-sample vulnerabilities, but that a single global threshold induces heterogeneous per-sample FPRs. This miscalibration can distort conclusions about which learning algorithms or defenses are more private (Section 4.1).
- To improve the concatenated approach, we propose a simple post-processing based approach that calibrates the per-sample FPRs and allows the concatenation to report average per-sample TPR with the targeted FPR under mild Gaussian assumptions (Sections 4.2 and 6).
- We demonstrate that the efficient LiRA approach by Carlini et al. (2022) suffers from finite population bias in estimating the in/out-distribution variances. This leads to a bias in the vulnerability evaluation (Sections 5 and 6).

2 Background

2.1 Membership inference attacks

The goal of the MIA is to infer whether a particular training sample belongs to the training dataset of a particular learning outcome Shokri et al. (2017). Denoting the target sample with x_{target} and the learning outcome with $\mathcal{M}_{\text{target}}$, MIA can be formally stated as the following binary hypothesis test

$$H_0 : x_{\text{target}} \notin D_{\text{train}} \quad \text{vs.} \quad H_1 : x_{\text{target}} \in D_{\text{train}}, \quad (1)$$

based on observing $\mathcal{M}_{\text{target}} \leftarrow \mathcal{A}(D_{\text{train}})$ where D_{train} is latent. In this work we will focus on *shadow-model* based MIA where the test statistic for the above hypothesis test is based on comparing the observed $\mathcal{M}_{\text{target}}$ to the distributions learning outcomes when x_{target} is in or out of the D_{train} . For a data distribution $\mathcal{P}$ the in and out samples are distributed as

$$D \sim \mathcal{P}^N \quad (2)$$

$$\mathcal{M}_{\text{out}} = \mathcal{A}(D \setminus \{x_{\text{target}}\}) \quad (3)$$

$$\mathcal{M}_{\text{in}} = \mathcal{A}(D \cup x_{\text{target}}). \quad (4)$$

We will denote the in and out distributions for a sample x respectively as $p_{\text{in},x}$ and $p_{\text{out},x}$.

The LiRA algorithm by Carlini et al. (2022) has proved to be a highly effective. From the trained shadow models, LiRA infers normal approximations for the in/out-distributions for each target point

separately. The LiRA score for a target point x and model $\mathcal{M}_{\text{target}}$ is given as

$$\mathcal{L}(x, \mathcal{M}_{\text{target}}) := \log \frac{p_{\text{in},x}(\mathcal{M}_{\text{target}})}{p_{\text{out},x}(\mathcal{M}_{\text{target}})} = \log \frac{\mathcal{N}(\mathcal{M}_{\text{target}}; \mu_{\text{in},x}, \sigma_{\text{in},x}^2)}{\mathcal{N}(\mathcal{M}_{\text{target}}; \mu_{\text{out},x}, \sigma_{\text{out},x}^2)}, \quad (5)$$

where the parameters $\mu_{\text{in},x}, \mu_{\text{out},x}, \sigma_{\text{in},x}, \sigma_{\text{out},x}$ are learned from the shadow models.

2.2 Variables in membership inference attacks

The success of MIA is affected by various factors arising both from the training algorithm as well as the data distribution.

Learning algorithm: We consider the learning algorithm to be a fixed program, however we allow the algorithm still to induce stochasticity into the learning outcome e.g. through subsampling of the training data or through operations such as dropout (Srivastava et al., 2014). Any stochasticity in the learning algorithm will affect the in/out-distributions, and therefore will have an effect on the MIA vulnerability.

Data distribution: The data distribution has a large effect on the success rate for the MIA. Since the in/out-distributions correspond to learning outcomes over different training sets drawn from the data distribution, it is easy to see that the MIA scores are affected by the data distribution.

Dataset level properties: Recently Tobaben et al. (2025) showed that the dataset level properties such as number of samples and number of classes can have an effect on the MIA vulnerability.

2.3 Measuring vulnerability

Since MIA is a binary hypothesis test, the vulnerability can be measured through the properties of the distinguisher performing this test. Carlini et al. (2022) suggested measuring the vulnerability in terms of the attack's TPR at low fixed FPR. This metric has been since widely adopted in the literature, see e.g. Zarifzadeh et al. (2024); Liu et al. (2022). Empirically, the FPR and TPR are commonly obtained from a ROC-curve built from the MIA classifiers' scores and corresponding ground truth labels.

Trade-off function. The FPR is an empirical estimates of a classifiers type I error, while the $\text{TPR} = 1 - \text{FNR}$ with FNR being the estimate of the type II error. *Trade-off function* gives a lower bound for any classifiers (FPR, FNR) curve, i.e. the inverse ROC-curve, and is defined as follows

Definition 2.1 (Dong et al. (2022)). For any distributions P and Q on a measurable space $(\mathcal{X}, \mathcal{F})$, the trade-off function $T(P, Q) : [0, 1] \rightarrow [0, 1]$ is given as

$$T(P, Q)(\alpha) = \inf\{\beta_\phi : \alpha_\phi \leq \alpha\} \quad (6)$$

where $\phi : \mathcal{X} \rightarrow [0, 1]$ is a rejection rule with

$$\alpha_\phi = \mathbb{E}_P[\phi], \quad \beta_\phi = 1 - \mathbb{E}_Q[\phi]. \quad (7)$$

The next Lemma shows that the trade-off function is invariant to bijective post-processing.

Lemma 2.1 (Bijective post-processing invariance). *Let P and Q be distributions on $\mathbb{R}$, and let $f(x)$ be a bijective function. Denote by P^f and Q^f the pushforward distributions of P and Q under f , i.e., $P^f(B) = P(f^{-1}(B))$ for all Borel sets $B \subseteq \mathbb{R}$, and similarly for Q^f . Then, for all $\alpha \in [0, 1]$,*

$$T(P, Q)(\alpha) = T(P^f, Q^f)(\alpha).$$

Proof. According to Lemma 2.2, for any functions f and g , for any $\alpha \in [0, 1]$ we have

$$T(P, Q)(\alpha) \leq T(P^f, Q^f)(\alpha) \leq T(P^{g \circ f}, Q^{g \circ f})(\alpha), \quad (8)$$

where $(g \circ f)(x) = g(f(x))$. Now, if f is an bijection it is invertible and we can choose $g = f^{-1}$, which yields

$$T(P, Q)(\alpha) \leq T(P^f, Q^f)(\alpha) \leq T(P^{f^{-1} \circ f}, Q^{f^{-1} \circ f})(\alpha) = T(P, Q)(\alpha) \quad (9)$$

$$\Rightarrow T(P, Q)(\alpha) = T(P^f, Q^f)(\alpha). \quad (10)$$

□

Lemma 2.2 (Dong et al., 2022, Lemma 1). *For any distributions P and Q and a function f , for any $\alpha \in [0, 1]$ we have*

$$T(P^f, Q^f)(\alpha) \geq T(P, Q)(\alpha). \quad (11)$$

3 Vulnerability of a sample

Assume now that we have access to an oracle that can compute the LiRA score for any target point/model pair $(x_{\text{target}}, \mathcal{M}_{\text{target}})$. Our goal is to use the scores from the oracle to address the following binary hypothesis test

$$H_0 : x_{\text{target}} \notin D_{\text{train}} \quad \text{vs.} \quad H_1 : x_{\text{target}} \in D_{\text{train}}. \quad (12)$$

The above test is often empirically assessed with an ROC-curve using a threshold classifier to separate the in and out scores. However, the above hypothesis test does not specify whether we consider the test over different $\mathcal{M}_{\text{target}}$ or x_{target} , i.e. do we consider the vulnerability of a sample over different learning outcomes, or do we consider the vulnerability of a learning outcome over different samples. In this paper we will focus on the former, which we will call the *per-sample* vulnerability. By selecting the target samples carefully, per-sample vulnerability allows estimating the vulnerability of the most vulnerable samples which is important for reliable vulnerability evaluation (Aerni et al., 2024).

In the per-sample setting, we assume the data distribution $\mathcal{P}$ is fixed and we study the binary hypothesis test over learning outcomes $\mathcal{A}(D_{\text{train}})$ with $D_{\text{train}} \sim \mathcal{P}$. Typically in MIA, we do not directly study the membership based on the learning outcome $\mathcal{A}(D_{\text{train}})$ but based on some post-processed quantity such as the logits scores in classification tasks. We will use $\mathcal{S}(x_{\text{target}}, \mathcal{A}(D_{\text{train}}))$ to denote the statistic used for MIA, where $\mathcal{S}$ is a deterministic function.

Now, with the target sample fixed, our binary test can be rewritten as

$$H_0 : \mathcal{S}(x_{\text{target}}, \mathcal{A}(D_{\text{train}})) \sim p_{\text{out},x} \quad \text{vs.} \quad H_1 : \mathcal{S}(x_{\text{target}}, \mathcal{A}(D_{\text{train}})) \sim p_{\text{in},x}. \quad (13)$$

In other words, we are studying the distinguishability of x_{target} from the learning outcome over the stochasticity arising from any inherent randomness of $\mathcal{A}$ and from the latent training dataset $D \sim \mathcal{P}^n$ that may or may not include x_{target} .

In order to empirically assess this vulnerability, we query the oracle with multiple models trained with or without x_{target} and build the ROC-curve with the corresponding scores and labels. The in/out scores can be written as

$$\ell_{x_{\text{target}}}^{(\text{out})} := \mathcal{L}(x_{\text{target}}, \mathcal{A}(D)) \quad (14)$$

$$\ell_{x_{\text{target}}}^{(\text{in})} := \mathcal{L}(x_{\text{target}}, \mathcal{A}(D \cup x_{\text{target}})), \quad (15)$$

where $D \sim \mathcal{P}^n$. The FPR and TPR for a threshold classifier with a decision threshold $\tau \in \mathbb{R}$ become

$$\text{FPR}_x(\tau) = \Pr(\ell_x^{(\text{out})} > \tau) = \mathbb{E}_{D, \xi}[\mathbf{1}[\ell_x^{(\text{out})} > \tau]] \quad (16)$$

$$\text{TPR}_x(\tau) = \Pr(\ell_x^{(\text{in})} > \tau) = \mathbb{E}_{D, \xi}[\mathbf{1}[\ell_x^{(\text{in})} > \tau]] \quad (17)$$

where the $D \sim \mathcal{P}^n$ and ξ denotes any additional stochasticity in $\mathcal{A}$.

4 Reporting aggregates of the vulnerability

Consider now that we have MIA scores evaluated over M target models and N target data points. Each of the N samples x_{target} was either in our out of the training data for each of the M models. We represent these results with a matrix $\mathcal{G} \in \mathbb{R}^{M \times N}$. Each row of $\mathcal{G}$ represents per-model scores, and each column per-sample scores.

In order to build the $\mathcal{G}$, we need to train M models. Therefore in many practical settings, our evaluations are often limited to small number M . However, in order to evaluate the MIA vulnerability in the low FPR setting, we would need at least $2/\text{FPR}$ models to even reach the desired FPR. And even if we have enough models to meet this criteria, the estimated ROC-curve can still be highly uncertain. This uncertainty prevents us from reliably estimating the per-sample vulnerability. However, when we have access to evaluations over multiple target models and target points, we can aggregate over these results in order to estimate the vulnerability.

4.1 Concatenating the MIA scores

Concatenating MIA scores over multiple target models has been used in the past to learn aggregates of the vulnerability (Carlini et al., 2022; Murakonda and Shokri, 2020). In concatenated MIA evaluation, we are collapsing the MIA grid $\mathcal{G}$ into a single vector, and evaluating the vulnerability based on a ROC curve built on these raveled scores and their corresponding labels.

The FPR and TPR for the concatenated approach are given as

$$\text{FPR} = \Pr(\ell^{(\text{out})} > \tau), \quad \text{TPR} = \Pr(\ell^{(\text{in})} > \tau), \quad (18)$$

where the dependence of both the model and sample to the scores is marginalized away. The next proposition demonstrates how the concatenated approach is related to the per-sample vulnerability.

Proposition 4.1. *For any fixed threshold τ , the concatenated FPR and TPR are equal to the expectation of per-sample FPR_x and TPR_x .*

$$\text{FPR}(\tau) = \mathbb{E}_x[\text{FPR}_x(\tau)], \quad \text{TPR}(\tau) = \mathbb{E}_x[\text{TPR}_x(\tau)].$$

Proof. For the concatenated FPR we have

$$\text{FPR} = \Pr(\ell^{(\text{out})} > \tau) = \mathbb{E}_{x,D,\xi}[\mathbf{1}[\ell_x^{(\text{out})} > \tau]] = \mathbb{E}_x[\mathbb{E}_{D,\xi}[\mathbf{1}[\ell_x^{(\text{out})} > \tau]]] = \mathbb{E}_x[\text{FPR}_x], \quad (19)$$

which similarly follows for the TPR. $\square$

Thus, choosing a single decision threshold τ that yields $\text{FPR} = \alpha$ at the concatenated level corresponds to choosing τ such that the average per-sample FPR_x equals α , while individual FPR_x may vary widely. Hence calibrating the concatenated FPR does not necessarily serve as a good proxy of understanding the vulnerability.

4.2 Post-processing for better aggregation

Next, we present our solution which solves the FPR calibration problem of the concatenated approach. For a fixed sample x , suppose the learning outcome $S(x, A(D))$ satisfies

$$S_x^{\text{out}} \sim \mathcal{N}(\mu_{\text{out},x}, \sigma_{\text{out},x}^2), \quad S_x^{\text{in}} \sim \mathcal{N}(\mu_{\text{in},x}, \sigma_{\text{in},x}^2), \quad (20)$$

and define the mean gap $\Delta_x = \mu_{\text{in},x} - \mu_{\text{out},x}$.

Now, consider the affine transform

$$f_x(t) = \frac{\text{sign}(\Delta_x)(t - \mu_{\text{out},x})}{\sigma_{\text{out},x}}. \quad (21)$$

Under this transform we obtain the standardized in/out distributions

$$\hat{S}_x^{\text{out}} := f_x(S_x^{\text{out}}) \sim \mathcal{N}(0, 1), \quad (22)$$

$$\hat{S}_x^{\text{in}} := f_x(S_x^{\text{in}}) \sim \mathcal{N}\left(\frac{|\Delta_x|}{\sigma_{\text{out},x}}, \frac{\sigma_{\text{in},x}^2}{\sigma_{\text{out},x}^2}\right). \quad (23)$$

By Lemma 2.1, this standardization does not change the trade-off function:

$$T(\mathcal{N}(\mu_{\text{in},x}, \sigma_x^2), \mathcal{N}(\mu_{\text{out},x}, \sigma_x^2))(\alpha) = T\left(\mathcal{N}\left(\frac{|\Delta_x|}{\sigma_{\text{out},x}}, \frac{\sigma_{\text{in},x}^2}{\sigma_{\text{out},x}^2}\right), \mathcal{N}(0, 1)\right)(\alpha). \quad (24)$$

4.2.1 Equal variances

Now, assume that $\sigma_{\text{in},x} = \sigma_{\text{out},x} := \sigma_x$. In this case from (24) we can see that all per-sample in/out pairs can be viewed as testing between a standard normal and a shifted unit-variance normal with mean $|\Delta_x|/\sigma_x$.

Next Lemma shows us that for the equal variance case, the likelihood-ratio is a bijection.

Lemma 4.1. *For two Gaussians $P = \mathcal{N}(\mu_1, \sigma^2)$ and $Q = \mathcal{N}(\mu_2, \sigma^2)$ with density functions p and q respectively. If $\mu_1 > \mu_2$ the log-likelihood ratio, $\Lambda(t) = \log \frac{p(t)}{q(t)}$ is a bijection w.r.t. $t \in \mathbb{R}$. Moreover, if $\mu_1 > \mu_2$, the $\Lambda(t)$ is also monotonically increasing.*

Proof. With simple algebra we can see that

$$\Lambda(t) = \frac{1}{2\sigma^2} (2t(\mu_1 - \mu_2) - \mu_1^2 + \mu_2^2). \quad (25)$$

Clearly $\Lambda(t)$ is an affine mapping and thus a bijection. $\square$

By Lemma 4.1, any level- α likelihood ratio test between P and Q can be implemented directly in the sample space: the Neyman–Pearson test that rejects Q when $\Lambda(t) > \tau_\alpha$ is equivalent to rejecting when $t > t_\alpha$, where t_α is the $(1 - \alpha)$ -quantile of the out-distribution under Q . In particular, since $|\Delta_x|/\sigma_x > 0$ the LLR is monotonically increasing in the learning outcome, so we can choose the decision threshold from the out-distribution and apply it directly to the observed value.

Since the out distribution for the post-processed learning outcomes is fixed to the standard normal among all the samples, we can use the same decision threshold $t_\alpha = \Phi^{-1}(1 - \alpha)$ to obtain the same FPR_x for all $x \sim \mathcal{P}$. Finally, we can estimate the average TPR as

$$\overline{\text{TPR}} = \frac{1}{NM} \sum_{\substack{i=1 \dots N \\ m=1 \dots M}} \mathbf{1}[\hat{S}_{i,j} > t_\alpha]. \quad (26)$$

Now, similar to the concatenated approach, our post-processed aggregated LiRA vulnerability gives an estimate of the vulnerability marginalized over both the models and sample in $\mathcal{G}$. Therefore, the metric is not suitable for assessing the per-sample nor the per-model vulnerability, but should be seen as a measure of how much the learning algorithm leaks about the individuals on average. However, as we are averaging over NM values in total, we can actually have a rather confident estimate of the underlying average per-sample vulnerability, even in the low FPR regime.

4.2.2 Unequal variances

When $\sigma_{\text{in},x} \neq \sigma_{\text{out},x}$, the LLR is no longer a bijection, which means that the rejection region for the LLR test cannot be uniquely mapped into a single sided test on the tails of the learning outcomes. Therefore testing if $\hat{S}_{i,j} > t_\alpha$ does not satisfy the conditions of Neyman-Pearson Lemma, and the test will have less power than the LLR test.

However, even though the test is not uniformly most powerful like the log-likelihood ratio test, we still retain calibration in terms of the per-sample FPRs. This is because the transformed learning outcomes $\hat{S}_x^{\text{out}}$ will follow the standard normal regardless. Therefore, the TPR arising from the concatenated transformed test will lower bound the true average vulnerability.

5 Efficient LiRA and Finite Population Bias

When implementing a MIA in practice on a real dataset, we do not have access to the underlying data generating process $\mathcal{P}$ but only a finite superset $D_{\text{full}} = \{x_1, \dots, x_{N_+}\} \sim \mathcal{P}^{N_+}$ from which the training sets of each model must be sampled. For each target model $\mathcal{M}$, the corresponding training set D of N elements is sampled without replacement from D_{full} .

Given these models, the “efficient LiRA” procedure proposed by Carlini et al. (2022) recycles these models so that each acts as a target model in turn, while the rest act as shadow models.

This yields an $M \times N_+$ LiRA grid analogous to our generic grid G , but with two important differences compared to the “ideal” LiRA setting where shadow datasets are independently drawn as $D \sim \mathcal{P}^N$:

- All shadow and target models are trained on overlapping subsets of the *same* finite D_{full} .
- Each D is sampled without replacement from D_{full} , rather than from an infinite population.

As a consequence, estimators of the in/out variances of the learning outcomes (or LiRA scores) are subject to a finite population bias. Intuitively, reusing the same finite set D_{full} multiple times makes the empirical variability of the shadow models smaller than what one would observe if each training set were drawn independently from $\mathcal{P}$.

5.1 Finite Population Correction

To see the effect at a high level, consider any scalar statistic $T(D)$ of a dataset D of size N drawn *without replacement* from a finite population of size N_+ . Classical sampling theory (Cochran, 1977) shows that

$$\text{Var}[T(D)] \approx \text{Var}_{\text{iid}}[T(D)] \cdot \left(1 - \frac{N}{N_+}\right), \quad (27)$$

where $\text{Var}_{\text{iid}}[T(D)]$ is the variance that would be obtained if the elements of D were sampled independently from the underlying population (i.e., with replacement). The factor

$$\text{FPC} = 1 - \frac{N}{N_+} \quad (28)$$

is the standard finite population correction. When N/N_+ is small, the correction is negligible, but for larger N the effect becomes significant. In efficient LiRA, it is common to choose N to be a large fraction of N_+ (e.g. $N = 0.5 N_+$), in which case $\text{FPC} \approx 0.5$ and the variance may be substantially underestimated.

In our setting, the statistic is the learning outcome (or its LiRA score) for a fixed point x across shadow models. Efficient LiRA estimates the in/out distributions for x from $\{\mathcal{M}^{(m)}\}_{m=1}^M$, but all $\mathcal{M}^{(m)}$ are trained on overlapping subsets of a fixed finite D_{full} . This induces an analogous finite population effect: the empirical in/out variances estimated from the shadow models under efficient LiRA are biased towards *smaller* values compared to the variances one would obtain from independent draws $D \sim P^N$.

Since LiRA’s distinguishability is driven in large part by the separation between the in and out distributions relative to their variances, underestimating these variances can lead to overly optimistic estimates of membership inference vulnerability. Moreover, the vulnerability estimated via efficient LiRA is tied to the particular realization D_{full} , rather than to the population-level distribution P , introducing an additional dataset-dependent bias.

6 Experiments

Experimental Setup For Efficient LiRA

- **Model** We use TabPFN (Hollmann et al., 2025), a model which leverages in-context learning (Garg et al., 2022) to predict labels for hitherto unseen training samples.
- **Dataset** We use 10K samples from UCI’s Adult (Becker and Kohavi, 1996) dataset and 1K samples from UCI’s Credit (Hofmann, 1994) dataset for binary classification tasks.

6.1 Effect of Post-Processing On Efficient LiRA Evaluation

In Fig. 1, we study following MIA evaluation strategies for a MIA grid of TabPFN models trained on Adult and Credit datasets:

- **Concatenated TPR (Naive):** All observations from the MIA grid are concatenated and the TPR is computed at a fixed FPR directly from the pooled statistics. This serves as a baseline that does not provide valid global FPR calibration.
- **Concatenated TPR (PP):** Before concatenation, the statistics in the MIA grid are post-processed as described in Section 4.2. The TPR at a fixed FPR is then computed from the post-processed, pooled statistics.
- **Average TPR/Sample:** The per-sample TPR, TPR_x , is computed separately for each sample in the MIA grid and averaged. By construction, this approach ensures that the per-sample $\text{FPR}_x \approx \alpha$ for every sample.
- **Concatenated TPR (PP, Normal):** The post-processed statistics are concatenated and the decision threshold is set analytically by modeling the out-distribution as $\mathcal{N}(0, 1)$ as in Section 4.2
- **Concatenated TPR (PP, Student’s t):** The decision threshold is set analytically using Student’s t -distribution with degrees of freedom estimated from the out-distribution. This yields a more conservative classifier that can account for heavy-tailed data.
- **Average TPR/Sample (Normal):** The per-sample TPR, TPR_x , is computed for each sample, but with the decision threshold calibrated analytically under the $\mathcal{N}(0, 1)$ assumption for the out-distribution.

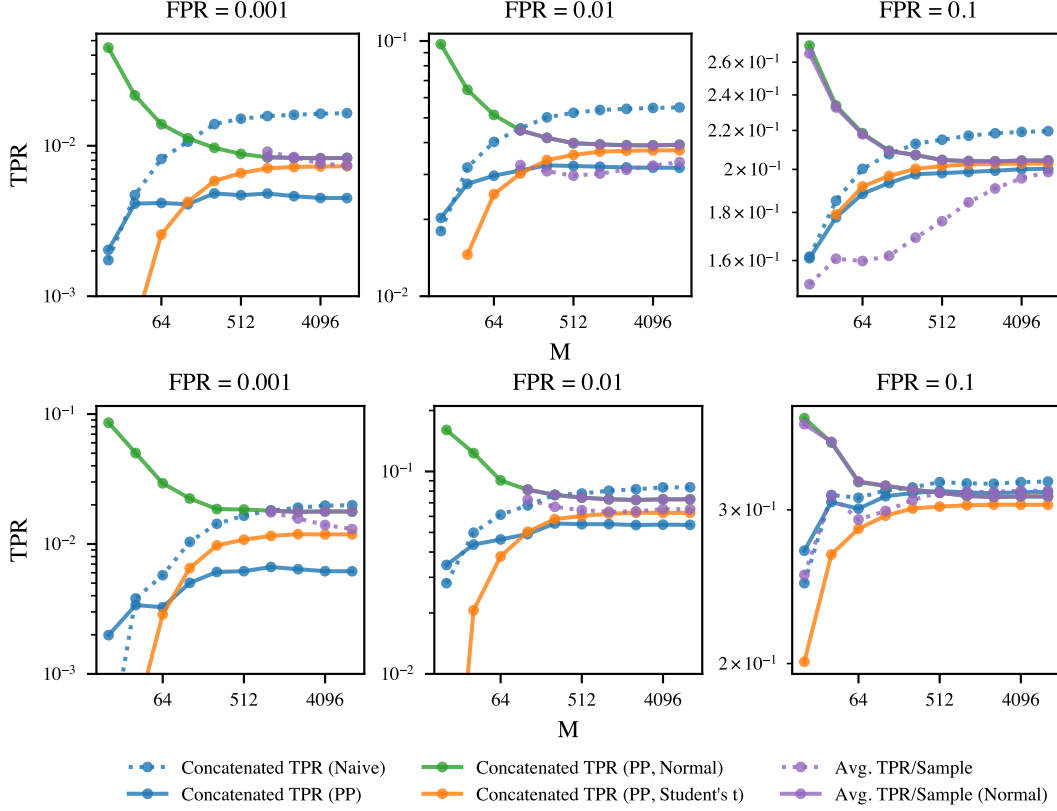


Figure 1: **Comparison of MIA evaluation strategies across varying M and FPRs** for TabPFN models trained on UCI Adult (top), and UCI Credit (bottom) datasets. Naive concatenation consistently yields higher TPRs, overestimating average privacy risk relative to post-processed concatenation which produces lower and more stable estimates. At small M , Average TPR/Sample is noisy; as M increases, it converges toward Concatenated TPR (PP), indicating that the latter is a conservative, but reliable finite-sample proxy for aggregated per-example privacy risk.

In Fig. 1, we observe that naive concatenation-based MIA evaluation consistently overestimates the average privacy risk of the training algorithm. Post-processing normalizes the per-sample out distributions before aggregation, ensuring that the decision threshold is applied on a common scale across all samples. Consequently, concatenated TPR (PP) is less than concatenated TPR (Naive) and this difference stabilizes at large M . Furthermore, we observe that the per-sample TPR estimates are noisy at small M because samplewise in/out distributions are estimated from fewer shadow models, whereas concatenation pools information but tends to suppress example-level heterogeneity. As M grows, Average TPR/Sample converges to Concatenated TPR (PP) showing that the latter acts as a finite-sample conservative approximation to Average TPR/Sample whose stability is contingent on large M . Using analytical decision threshold from $\mathcal{N}(0, 1)$ leads to inflated TPR estimates, especially when $M \leq 512$. Using the decision threshold from Student’s t-distribution can yield accurate results when $M \geq 256$.

Fig. A2 shows the distribution of per-sample FPRs across samples for multiple target FPRs ($\alpha \in \{0.001, 0.01, 0.1\}$) computed using $M = 4096$ models. The naive concatenated approach produces FPR distributions with heavy tails indicating a non-trivial fraction of samples get assigned an $\text{FPR} > \alpha$. It does not account for different samplewise score variances. Instead, it applies a single global decision threshold across all samples. Meanwhile, the FPR_x for evaluations using post-processed statistics bring these distributions closer to the target FPR.

In Section A.1, we consider an oracle-access setting where the attacker uses the MIA grid for estimating the per-sample in/out distributions. Relying on this assumption, we can isolate the effect of the evaluation strategy itself. This allows us to compare Concatenated TPR (Naive), Concatenated TPR (PP) and Average TPR/Sample without confounding from finite- M samplewise estimation error. Consistent with the results in Fig. 1, Average TPR/Sample tends to converge to Concatenated TPR (PP) for large M showing that the latter is an efficient and cheap proxy for the former.

6.2 Analytical Simulation of Efficient LiRA

To demonstrate the finite population bias on efficient LiRA evaluation, we consider D_{full} to be sampled from a multivariate Gaussian distribution, $x \sim \mathcal{N}(0, \sigma^2 \mathbb{I}_d)$. We define the model outcome $\mathcal{M}^{(m)}$ as:

$$\mathcal{M}^{(m)} = \frac{1}{|D^{(m)}|} \sum_{x \in D^{(m)}} x, \quad D^{(m)} \subset_N D_{\text{full}}.$$

Given a target pair $(x_i, \mathcal{M}^{(m)})$, we use its inner product with the empirical mean as the target statistic:

$$s_i^{(m)} = \langle x_i, \mathcal{M}^{(m)} \rangle.$$

Figure Fig. 2a demonstrates that using a finite superset and a finite M allows for empirical standard deviation (σ_{emp}) to be consistently smaller compared to the expected analytical standard deviation per-sample (σ_{ana}). Applying the FPC (Eq. (28)) helps scale up σ_{emp} close to σ_{ana} . Figure Fig. 2b further shows the distribution of $\frac{\sigma_{\text{emp}}}{\sigma_{\text{ana}}}$ across D_{full} for both in/out distributions, estimated from $M = 2048$ shadow models fitted on a finite superset of size $N_+ = 1000$ with $N = 500$, and $d = 500$. The ratios concentrate around $\sqrt{\text{FPC}}$ as predicted in Section 5.1. This confirms that efficient LiRA’s finite population bias arises from subsampling training datasets for shadow models from a finite superset D_{full} . Not accounting for FPC leads to variance miscalibration that grows with the sampling ratio N/N_+ (see Fig. A3).

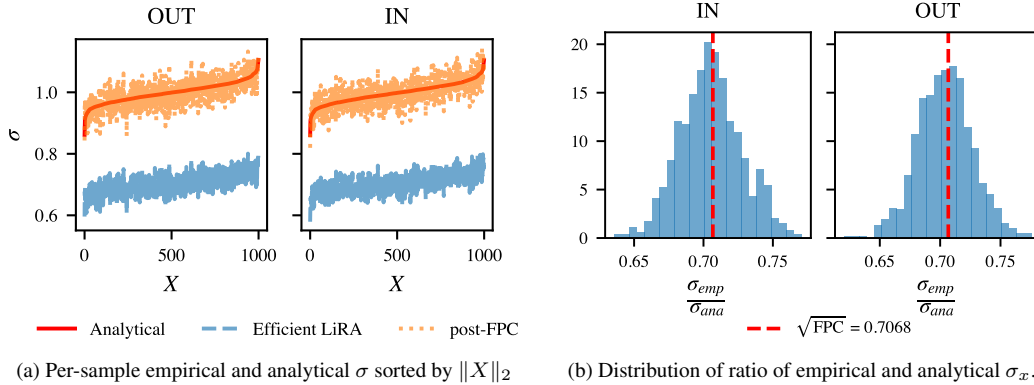


Figure 2: **FPC correction can help recover the analytical σ .** (a) depicts empirical σ with and without FPC. We observe that post-FPC σ_x closely track the expected analytical σ_x . In (b), we observe that FPC recovers the analytical σ_x but with some residual scatter as depicted by the spread of $\sigma_{\text{emp}}/\sigma_{\text{ana}}$ around $\sqrt{\text{FPC}}$. The plots use data from running the simulation described in Section 6.2 with $M = 2048$, $N_+ = 1000$, $d = 500$, and $N = 500$.

7 Conclusion

Our results show how to evaluate the average MIA vulnerability across a number of data points in terms of TPR at fixed low FPR more reliably and efficiently. While this is a quantity we can evaluate efficiently and is clearly linked to the vulnerability, it is not ideal for this purpose. As pointed out by Aerni et al. (2024), vulnerability evaluation should focus on the most vulnerable data points, which can be watered down in the average. This means that the average TPR values need to be interpreted with care. However, any plausible alternative would require being able to identify the most vulnerable data points, and training a large number of models while including/excluding those points, which is highly non-trivial. Our approach provides a relatively cheap and easy alternative for evaluating MIA vulnerability as a valid TPR at known FPR which is not available from previous methods. Source code for our experiments is available at https://github.com/TrustworthyMLHelsinki/mi_vulnerability_evaluation.

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A Appendix

A.1 MIA Evaluation With Oracle Access

In this setting, the attacker uses all 4095 models except the target model to compute per-sample statistics for S_x^{out} and S_x^{in} against M target models. It helps isolate the behavior of the MIA evaluation procedure from finite- M estimation error of the in- and out-distributions. In particular, it lets us study how naive concatenation, post-processed concatenation, and average per-sample TPR relate when the samplewise distributions are estimated accurately. Thus, the oracle plots provide an idealized reference for the privacy-relevant quantity that efficient LiRA variants aim to approximate. However, it should be interpreted as a calibration benchmark for the evaluation methods, not as a claim about the capabilities of a compute-limited attacker.

As shown in Fig. A1, under oracle access, Concatenated (Naive) approach overestimates average privacy risk, while Concatenated TPR (PP) provides a calibrated, conservative aggregate approximation to the per-sample benchmark. As with Fig. 1, Average TPR/Sample computations converges towards Concatenated TPR (PP) as M increases, suggesting that the post-processed concatenated statistic remains a strong finite-sample approximation to average per-sample privacy risk.

A.2 Additional Plots

A.3 Licenses and Access

- **UCI Adult** (Becker and Kohavi, 1996): This dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license and we use the dataset available at <https://doi.org/10.24432/C5XW20>.
- **UCI German Credit Data** (Hofmann, 1994): This dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license and we use the dataset available at <https://doi.org/10.24432/C5NC77>.

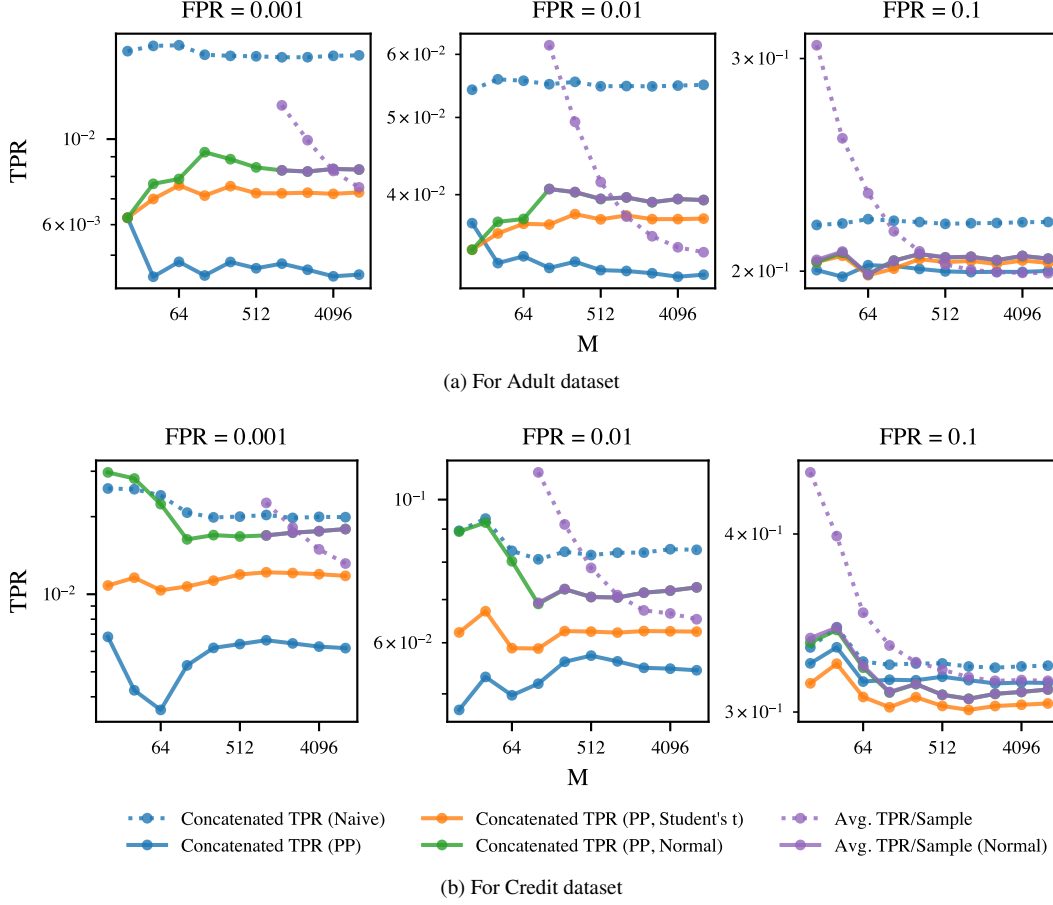


Figure A1: **Oracle-access MIA evaluation.** The MIA grid is used to compute per-sample statistics isolating effect of finite- M samplewise estimation error. Following that, we compare different MIA evaluation strategies across varying M and FPRs for TabPFN models trained on (a) UCI Adult, and (b) UCI Credit datasets. Variation in M Naive concatenation overestimates average privacy risk, while Concatenated TPR (PP) provides a calibrated, conservative aggregate approximation to Average TPR/Sample.

- **TabPFN** (Hollmann et al., 2025): We use TabPFN model hosted on HuggingFace at https://huggingface.co/Prior-Labs/tabpfn_2_5. It is licensed under a non-commercial Prior Labs License (Apache 2.0).

A.4 Computational Resources

We train the TabPFN model on NVIDIA Volta V100 GPUs (with 32 GB VRAM) and 10 GB of host memory. Precise estimates for compute for training time are hard to make as it varies by task difficulty and dataset size. Though, to give a rough estimate of scale, training a single TabPFN model on Adult (with $\approx 5K$ samples) takes < 2 minutes with a GPU.

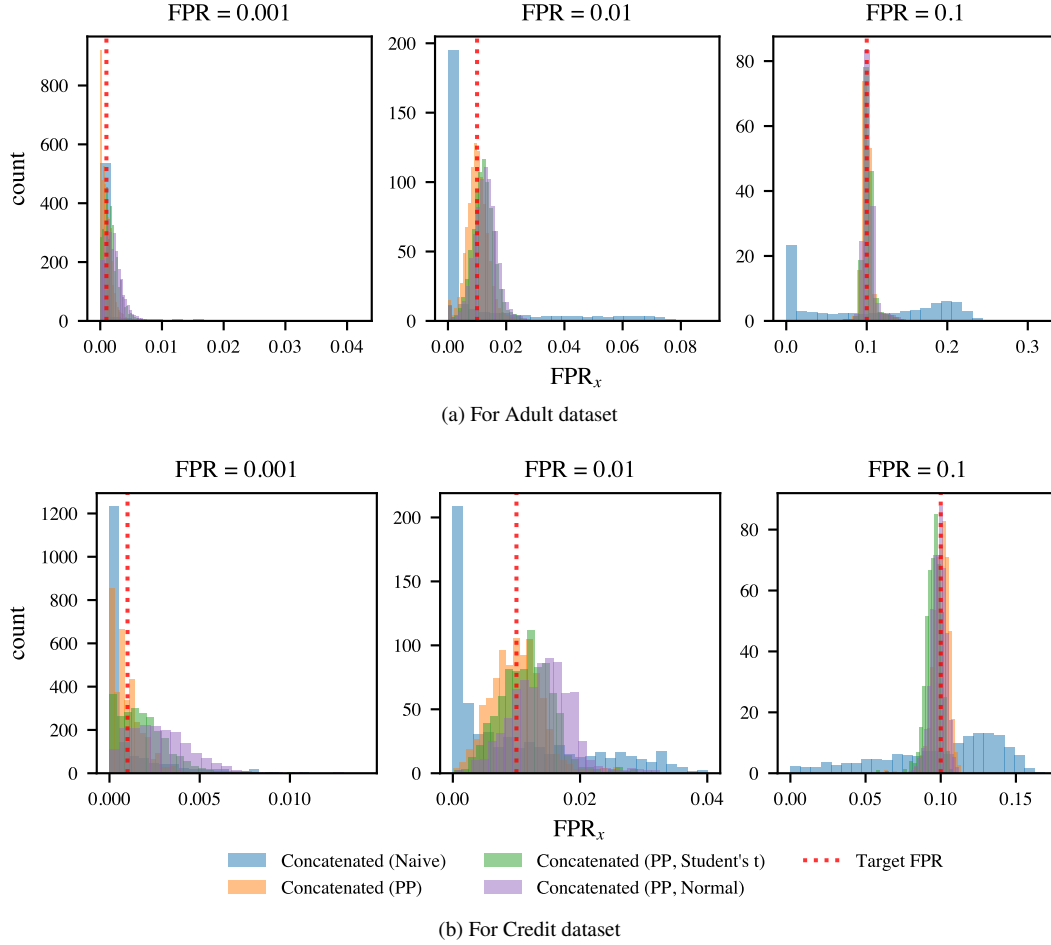


Figure A2: **Distribution of per-sample FPRs evaluated at decision thresholds for various concatenation-based MIA evaluation strategies.** Naive concatenation applies single decision threshold to unnormalized statistics, leading to uneven FPRs across samples. Post-processing normalizes samplewise out distributions before aggregation, improving calibration and reducing per-sample FPR heterogeneity. The plots are generated for a MIA grid of TabPFN models fitted on Adult and Credit datasets.

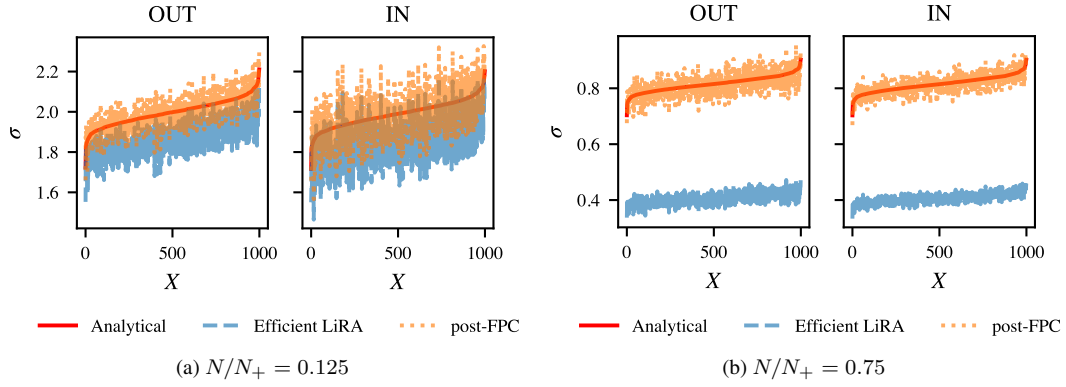


Figure A3: **Finite Population Bias depends on subsampling ratio (N/N_+).** The plots depict the difference between analytical and empirical standard deviation (σ_x) as a factor of subsampling ratio (N/N_+). As $N/N_+ \rightarrow 0$, empirical σ_x computed over the MIA grid ceases to be a good estimate of analytical sigma. FPC correction (Section 5.1) helps to scale up empirical σ_x to their analytical equivalent. The plots use data from running the simulation described in Section 6.2 with $M = 2048$, $N_+ = 1000$, $d = 500$, and varying N .